MILITARY SPECIFICATION

LUBRICATION CHARTS FOR AIR WEAPON SYSTEMS, PREPARATION OF

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers the preparation of maintenance lubrication charts for aircraft and associated ground support equipment.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Bureau of Naval Weapons Instructions

BuWeps Instruction 10300.1	Aviation fuels, lubricants and specialty products used by the NATO Armed Forces
BuWeps Instruction 4700.2	The Naval Aircraft Maintenance

(Copies of BuWeps Instructions may be obtained from the Supply Department, NAVSTA (NAVYRD Annex, Code 541.25) Washington, D. C. 20360)

3. REQUIREMENTS

3.1 General Remarks

3.1.1 Data - The lubrication charts furnished under this specification shall show lubrication requirements for all parts requiring periodic lubrication. Charts shall cover systems, assemblies or components logically grouped together to show an easily understood and orderly presentation of lubrication requirements.

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3.1.2 Content - Each chart requires a main and necessary supplemental drawings, prepared as diagrammatic outlines, indicating all symbols, keys, legends, nomenclature and other information that will clearly and concisely show the type of lubricant, lubrication interval and method of application. Charts shall convey fully to the user the complete lubrication requirements without referring to supplemental material in the parent manual or other documents. Should deviations be required, then all information of a general nature, such as keys, symbols, and explanatory notes, shall be shown on a page preceding the chart(s). The text of the chart shall be brief and explicit, providing the necessary and special instructions to be observed in the use of the chart. Supplemental drawings, showing enlarged and detailed views, shall be used to identify otherwise obscured lubricated items.

3.1.3 Arrangement - The general arrangement and format of the lubrication chart shall conform to Figure 4, Sheets 1, 2, and 3. Detailed specification governing the organization and content of the charts are set forth in Sections 3.2 and 3.3.

3.1.4 Application - The provisions of this section and section 3.2 and 3.3 apply equally to the preparation of master lubrication charts and maintenance requirement cards (MRC) except that an interval symbol key list is not required on the MRC.

3.2 Chart format -

3.2.1 Lubricating symbols - A lubrication symbol shall consist of a NATO, or a special lubrication number, an application symbol, and a parts nomenclature number, as required, in that order from top to bottom, enclosed in an interval symbol. Special note numbers may be included when necessary.

3.2.2 Symbol application - The lubrication symbol shall have a leader line extending to the point on the diagram, or drawing, that requires servicing. The symbol shall designate one point of servicing; however, several leader lines may branch from the main leader to designate adjacent points of servicing provided the following conditions are satisfied:

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a. The points of servicing are located on the same or related parts having the same nomenclature.

b. The points of servicing appear in the same view on the diagram or drawing.

c. The points of servicing require identical service.

A point of servicing which has been designated by a lubrication symbol in one view shall not be redesignated in another view. The points of servicing shall be designated in the view showing it more clearly.

3.2.2.1 Alternate method - Lubrication charts not requiring part nomenclature keys, see 3.2.4.1, and containing a large number of points of servicing having the same lubrication symbol, may be designated with numbered leader lines as shown in Figure 4, Sheet 2. The lubrication requirements shall be called out with a lubrication symbol placed immediately above the note "XX Places, Numbers XX Through XX" or "XX Places (As Numbered)."

3.2.3 Symbol arrangement - Lubrication symbols on charts shall begin with the lowest parts nomenclature number and conform to one of the following arrangements:

a. Start at the upper left corner of the chart and progress numerically in a clockwise direction around the diagram or drawing.

b. Arranged in vertical rows from top to bottom.

c. Arranged in horizontal rows from left to right.

When a multi-sheet chart is used, the arrangement on each succeeding sheet need not correspond to the arrangement of the first sheet. Lubrication symbols shall be placed adjacent to, but not so close as to crowd the diagram or drawing.

3.2.4 Parts nomenclature key - Each chart shall contain a parts nomenclature key where every part requiring lubrication shall be given a sequential parts nomenclature number beginning with Number 1.

3.2.4.1 The requirements of 3.2.4 do not apply to charts of sufficient completeness and clarity to make further identity of items to be lubricated unnecessary.

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3.2.5 Table of Lubricants - Each chart shall have a three column Table of Lubricants listing the following information:

a. NATO Number, as defined in BUWEPS INST. 10300.1A, or special Number of the lubricant.

b. Nomenclature of the lubricant.

c. Specification applicable to the lubricant.

3.2.6 Interval and application symbols - Each chart shall have an interval and application symbol key explaining the symbols used.

3.2.7 Special notes - Special notes may appear on a chart, or sheet, when necessary to explain special circumstances not otherwise provided for by the standard symbols specified herein. Such notes shall be numbered in sequence, beginning with the Number 1, on each sheet.

3.3 Lubrication Symbols -

3.3.1 Interval symbols - Shapes and relative sizes of interval symbols shall conform to the dimensions given in Figure 1.

3.3.1.1 Interval symbols show when equipment shall be lubricated. The term "Periodic" is the "Intermediate" inspection interval defined in BUWEPS INST. 4700.2. For aircraft where no intermediate inspection is required (e.g. A5A and A6A) the "Periodic" shall be the "Major" inspection interval.

3.3.2 Application symbols - Shapes and relative sizes of application method symbols are shown in Figure 2. Sizes shall not be reduced to a point where legibility is affected. No other symbols are authorized.

3.3.3 Symbol designation - Parts nomenclature key and lubrication symbols shall be as specified in 3.2.4 and item (a) of 3.2.5. Lubricants not having assigned NATO numbers shall be designated by a number enclosed in a triangle. See figure 3 for examples and explanatory notes of symbols used.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, methods of inspection and approval shall be in accordance with those specified for the manual, handbook or other document containing the lubrication charts.

5. PREPARATION FOR DELIVERY

5.1 This section is not applicable to this specification.

6. NOTES

6.1 Definitions - The term "Maintenance Lubrication Charts" includes, but is not limited to, those charts used in Maintenance Instruction Manuals, Handbook of Inspection Requirements, Periodic Maintenance Requirements Manuals, Maintenance Requirements Cards (MRC), and similar controlling documents.

6.2 International Agreement - Certain provisions (see 3.2.1, 3.2.5 and 3.3.3) of this specification are the subject of international standardization Agreement (A.B.C. AIR STANDARDS 15/1E Fuels, Lubricants, and Liquid Propellants and NATOSTANAG 3437 Fuels, Lubricants, and Liquid Propellants.) When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

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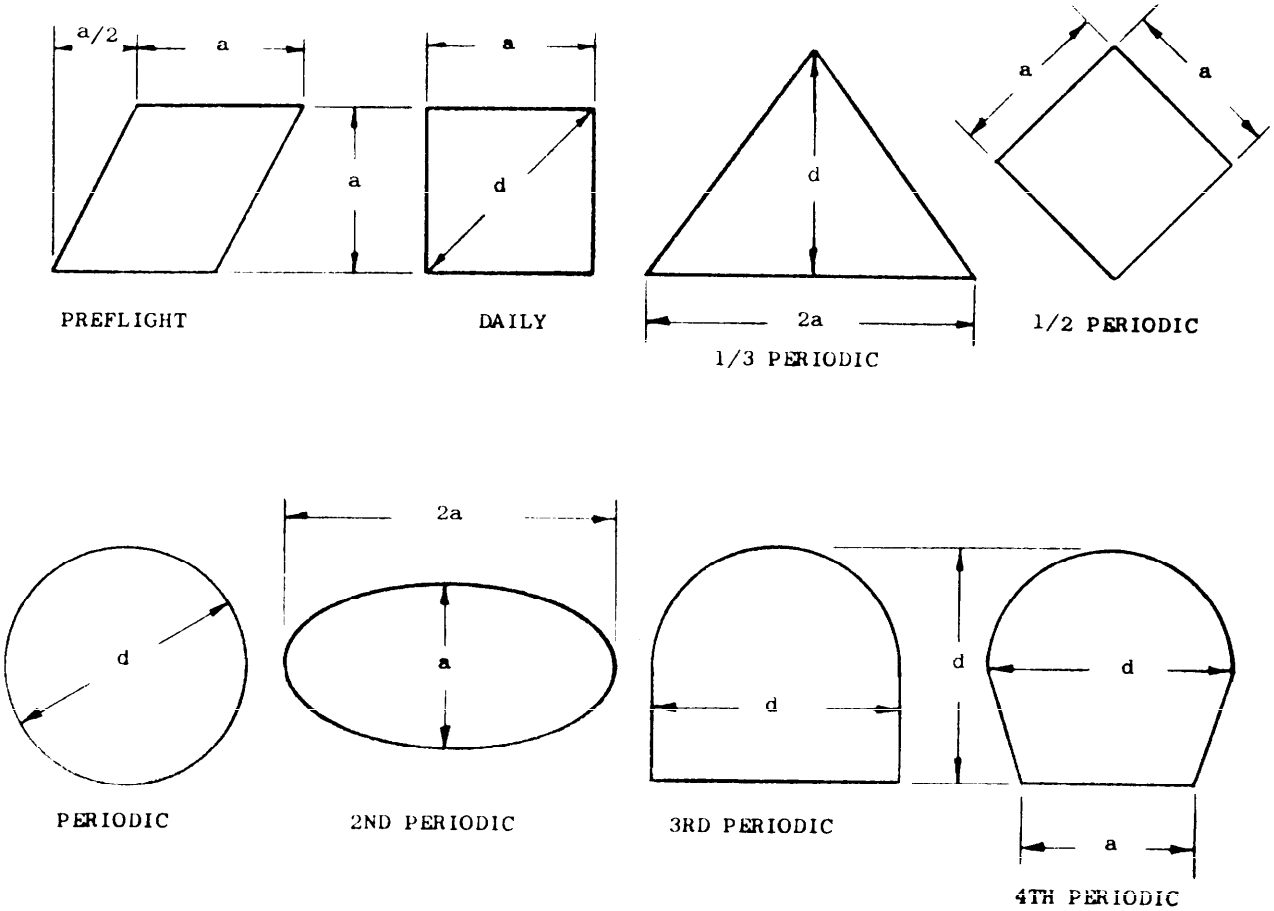


FIGURE 1, INTERVAL SYMBOLS

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ZERK
FITTING



MINIATURE
FITTING



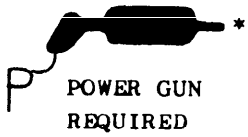
FLUSH
FITTING



RIGID
EXTENSION



FLEXIBLE
EXTENSION



POWER GUN
REQUIRED



SPECIAL
REQUIREMENT

LUBIRCATION GUN



OIL CAN



HAND



BRUSH



STICK

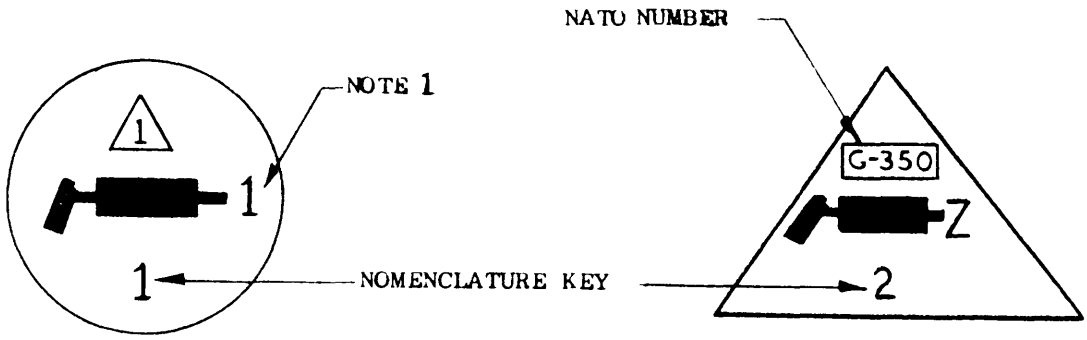
HYPODERMIC
NEEDLE

OTHER METHODS

* THE APPLICABLE FITTING SYMBOL WILL BE ATTACHED.
** A NOTE REFERENCE NUMBER WILL BE PLACED IN THIS POSITION. THE NOTE WILL EXPLAIN SPECIAL REQUIREMENTS.

FIGURE 2. APPLICATION METHOD SYMBOL

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EXAMPLE 1

EXAMPLE 2

TABLE OF LUBRICANTS		
NATO NUMBER	NOMENCLATURE	SPECIFICATION NO.
G-350	GREASE, AIRCRAFT AND INSTRUMENT	MIL-G-3278A
1	GREASE, SPECIAL PURPOSE FSN9150-274-0000	

PARTS NOMENCLATURE KEY
1. DRIVE BRAKE PISTON TERMINALS (L/R)
2. STABILIZER PIVOT POINT

EXAMPLE NO. 1 - APPLY SPECIAL PURPOSE GREASE, FSN9150-274-0000 WITH A LUBRICATION GUN USING MDE32447-1, 90° FITTING ASSEMBLY TO THE DRIVE BRAKE PISTON TERMINALS EVERY PERIODIC INSPECTION.

EXAMPLE NO. 2 - APPLY MIL-G-3278A GREASE WITH A LUBRICATION GUN HAVING A ZERK FITTING ADAPTER TO THE STABILIZER PIVOT POINT EVERY 1/3 PERIODIC INSPECTION.

NOTE:

1 USE MDE32447-1, 90° FITTING ASSEMBLY.

FIGURE 3, SAMPLE LUBRICATION SYMBOLS

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TABLE OF LUBRICANTS

NATO NUMBER	SPECIFICATION NUMBER	NOMENCLATURE
O-142	MIL-L-7870	LUBRICATING OIL, GENERAL PURPOSE, LOW TEMPERATURE
G-350	MIL-G-3278	GREASE, AIRCRAFT & INSTRUMENT (FOR LOW & HIGH TEMPERATURE)
G-353	MIL-G-21164	GREASE, MOLYBDENUM DISULFIDE (FOR LOW & HIGH TEMPERATURE)
G-361	MIL-G-25760	GREASE, AIRCRAFT, BALL & ROLLER BEARING, WIDE TEMP. RANGE
H-515	MIL-H-5606	HYDRAULIC FLUID, PETROLEUM, BASE, AIRCRAFT & ORDNANCE.
O-190	MIL-L-644	LUBRICATING OIL, GENERAL PURPOSE, PRESERVATIVE (WATER DISPLACING LOW TEMP.)

COMPONENT LUBRICATION SHEET INDEX

NO.	COMPONENT	SHEET NO.
1	PROBE, INFLIGHT FUELING	4
2	CANOPY, FORWARD	8
3	CANOPY, AFT	6
4	FINGERSEALS, AFT FUSELAGE	9
5	FLAPS, TRAILING EDGE	12
6	AILERONS	5
7	SPOILERS	7
8	WINGFOLD	10
9	FLAPS, LEADING EDGE	11
10	ELEVATOR AND RUDDER CONTROL	2
11	NOSE LANDING GEAR	3

SAMPLE NOTES

1. AREA INDICATED ON THIS ILLUSTRATION (ALL SHEETS) BY [illegible] ARE TYPICAL OF BOTH LEFT AND RIGHT HAND SIDES OF THE AIRPLANE.
2. LUBRICATE AZIMUTH-ELEVATION RANGE INDICATOR AND ANTENNA CONTROL MOUNTING CARRIAGE STOWAGE TRACKS WITH GREASE (MIL-G-7187) EVERY 30 DAYS (DAILY INSPECTION).

CAUTION

MIL-G-25760 IS NOT COMPATABLE WITH OTHER LUBRICANTS. WHEN IT IS SPECIFIED WITH USE IN AN AREA PREVIOUSLY LUBRICATED WITH A DIFFERENT GREASE, CLEAN AWAY THE OLD LUBRICANT BEFORE APPLYING THE ABOVE ITEM.

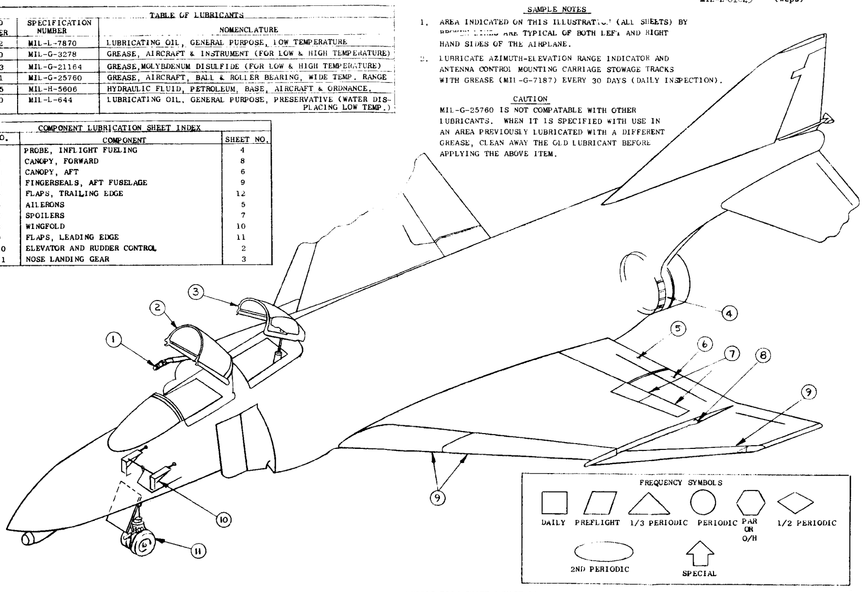
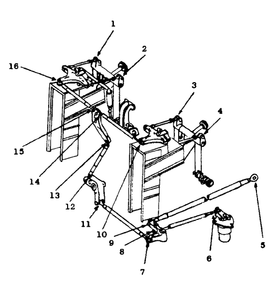


FIGURE 4, SAMPLE LUBRICATION CHART (SHEET-1)

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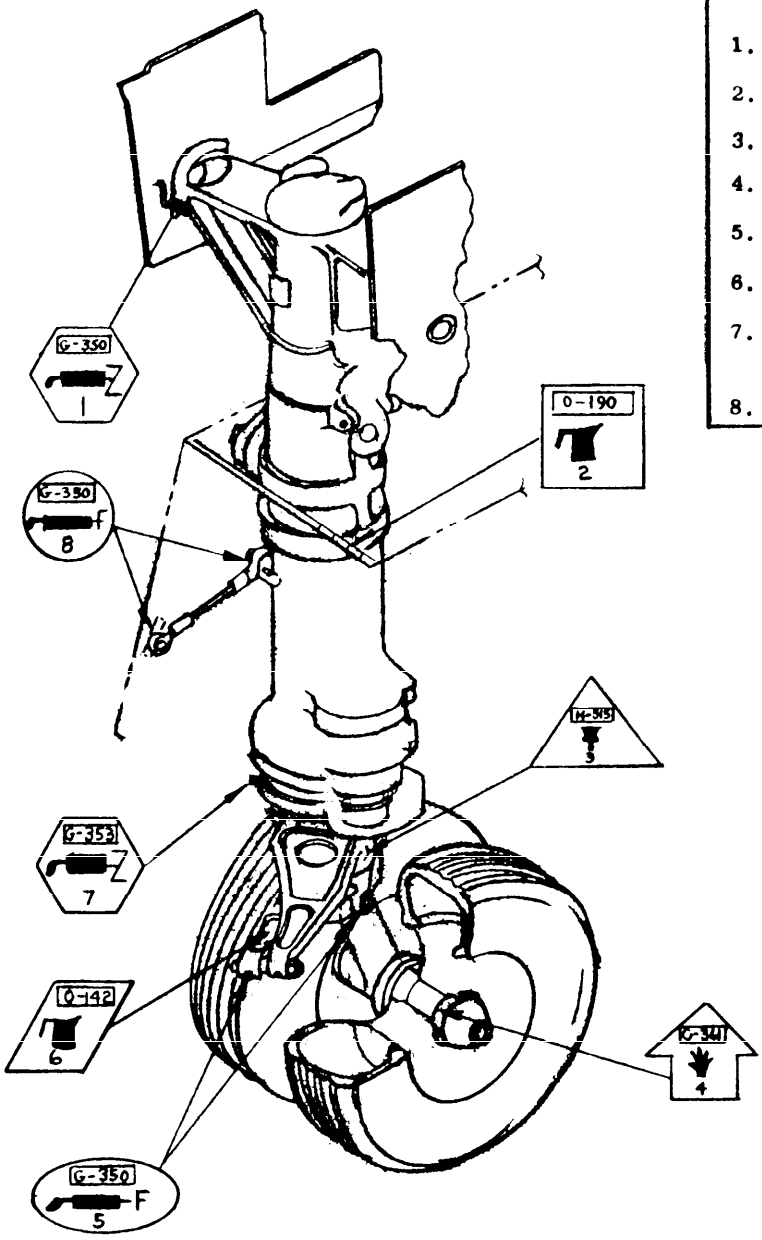
16 PLACES
(AS NUMBERED)



ELEVATOR AND RUDDER CONTROL

FIGURE 4. SAMPLE OF MULTIPLE LUBRICATION POINTS (SHEET 2)

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PARTS NOMENCLATURE

1. TRUNNIONS
2. HINGE PIN
3. SHOCK STRUT PISTON
4. WHEEL BEARINGS
5. TORQUE ARMS
6. JACK PAD
7. TORQUE COLLAR AND RING GEAR COVER
8. LINK ASSY, FORWARD DOOR

NOSE LANDING GEAR

FIGURE 4, SAMPLE LUBRICATION CHART (SHEET 3)

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SPECIFICATION ANALYSIS SHEET

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Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-L-81043(Wep)LUBRICATION CHARTS FOR AIR WEAPON SYSTEMS, PREPARATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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